TransportIDF — Réseau de Transport d'Île-de-France

Part 3: Maintenance, Incidents and Data Integration

Internal Document — Engineering & Data Governance Department

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1. Trip Operations and Real-Time Tracking

Trip Records

Every revenue service run is recorded as a **trip** in the operational database:

Field	Description

Trip ID	Unique trip identifier
Line ID	Which line the trip operates on
Vehicle ID	Assigned vehicle
Direction	Outbound or inbound
Trip Date	Date of service
Departure / Arrival Time	Scheduled start and end
Duration (minutes)	Total trip time
Status	Completed, cancelled, delayed, or in progress

With **1,500 trip records**, the dataset models a representative sample of daily operations.

Trip Status Distribution

Status	Share	Description
Completed	~75%	Trip ran as scheduled
Delayed	~12%	Trip ran with significant delay
Cancelled	~8%	Trip was suppressed
In Progress	~5%	Currently running (real-time snapshot)

Stop Events: Real-Time Precision

Each trip generates multiple **stop events** — one per station visited:

Field	Description
Stop Event ID	Unique identifier
Trip ID	Parent trip
Station ID	Station where the event occurred
Scheduled Arrival	Timetable arrival time
Actual Arrival	Real-time recorded arrival
Delay (seconds)	Difference between scheduled and actual

Passenger Boarding	Passengers entering at this stop
Passenger Alighting	Passengers exiting at this stop
Platform	Platform or quai identifier

With **3,000 stop event records**, this table provides the foundation for:

- **Punctuality metrics** — Percentage of stops served within 2 minutes of schedule
- **Load profiling** — Passenger counts by stop to identify crowding hotspots
- **Dwell time analysis** — Time spent at each station for schedule optimisation
- [redacted] — How delays cascade through the line

Real-Time Data Sources

Stop event data in production systems is fed by:

- **SIEL** (Système d'Information En Ligne) — Real-time passenger information
- **SAE** (Système d'Aide à l'Exploitation) — Operations management system
- [redacted] Régionale d'Information Multimodale)

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2. Incident Management

Incident Taxonomy

Network disruptions are classified into **12 incident types**:

Incident Type	Description	Typical Severity
Signal Failure	Signalling system malfunction	Major
Track Fault	Rail or track infrastructure defect	Major
Rolling Stock Issue	Vehicle breakdown or malfunction	Moderate
Passenger Incident	Medical emergency, altercation	Minor–Moderate
Power Outage	Traction power loss	Critical
Door Malfunction	Train door failure	Minor
Medical Emergency	Passenger requiring medical attention	Moderate
Suspicious Package	Security alert	Major–Critical

Strike Action	Industrial action by staff	Critical
Weather Disruption	Storm, flooding, extreme heat	Major
Trespasser on Track	Unauthorised person on tracks	Major
Overcrowding	Platform or vehicle capacity exceeded	Minor

Severity Levels

Severity	Impact	Response Time	Escalation
Minor	< 5 min delay, localised	< 15 min	Station controller
Moderate	5–30 min delay, line section	< 30 min	Line operations centre
Major	30–120 min, full line	Immediate	Network operations centre
Critical	> 2 hours, multiple lines	Immediate	Crisis management team

Incident Record Structure

Field	Description
Incident ID	Unique identifier
Line ID	Affected line
Station ID	Affected station (if applicable)
Incident Type	Classification
Severity	Minor, moderate, major, critical
Description	Free-text description
Start / End Datetime	Duration of the incident
Duration (minutes)	Total disruption time
Passengers Affected	Estimated impacted passengers
Status	Resolved, ongoing, or under investigation

Incident Statistics (Annual)

Metric	Value
Total incidents (dataset)	300
Average duration	~65 minutes
Most common type	Rolling stock issues (~18%)
Most common severity	Moderate (~35%)
Incidents on metro lines	~45% of total
Peak incident hours	07:00–09:00 and 17:00–19:00

With **300 incident records**, this dataset enables disruption pattern analysis, root cause investigation, and resilience planning.

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3. Fleet Maintenance

Maintenance Strategy

TransportIDF operates a **mixed maintenance strategy** combining:

1. **Preventive maintenance** — Scheduled inspections at fixed intervals
2. **Corrective maintenance** — Reactive repairs following breakdowns
3. **Predictive maintenance** — Data-driven anticipation of failures (emerging)

Maintenance Task Types

Task Type	Description	Avg. Cost (EUR)
Preventive Inspection	Regular scheduled check	500–2,000
Corrective Repair	Fix after failure	1,000–15,000
Overhaul	Major component replacement	10,000–50,000
Cleaning	Interior/exterior cleaning	50–300
Brake Check	Safety-critical brake inspection	200–800
Door Repair	Door mechanism repair	500–3,000

HVAC Service	Air conditioning maintenance	300–2,000
Software Update	Onboard systems firmware update	100–500
Wheel Profile	Wheel wear measurement and turning	2,000–8,000
Pantograph Check	Current collector inspection (rail)	1,000–5,000

Maintenance Record Structure

Field	Description
Task ID	Unique identifier
Vehicle ID	Which vehicle (if vehicle-related)
Station ID	Which station (if infrastructure-related)
Task Type	Category of maintenance
Priority	Low, medium, high, urgent
Scheduled Date	Planned maintenance date
Completed Date	Actual completion date
Cost (EUR)	Task cost
Technician ID	Assigned maintenance technician
Status	Completed, in progress, scheduled, cancelled

Maintenance Facilities

Facility	Location	Modes Served
Ateliers de Fontenay	Fontenay-sous-Bois	Metro (Lines 1, 14)
Centre de maintenance Bobigny	Bobigny	Metro (Lines 5, 11)
TMF Saint-Ouen	Saint-Ouen	Metro (Line 14 — MP14)
Centre bus Lagny	Paris 20e	Bus fleet
Ateliers de Villeneuve	Villeneuve-Saint-Georges	RER D

CIMT Sucy	Sucy-en-Brie	RER A (MI09, MI2N)

Maintenance KPIs

KPI	Target	Current
Fleet availability rate	> 95%	93.8%
Mean Time Between Failures (MTBF)	> 50,000 km (metro)	48,200 km
Preventive/corrective ratio	> 70/30	65/35
Maintenance cost per vehicle-km	< €0.15	€0.17
On-time task completion	> 90%	87%

With **200 maintenance task records**, this dataset supports fleet lifecycle management, cost analysis, and reliability engineering.

4. Data Model Summary

The TransportIDF data model comprises **16 tables** organised in four layers:

Reference Data

Entity	Description	Records
Transport Operator	Contracted transport companies	20
Transport Mode	Metro, RER, Bus, Tramway, etc.	8
Line	Individual service lines	55
Station	Network stations and stops	247
Ticket Type	Fare products	15

Network Structure

Entity	Description	Records
Line Station	Line-to-station mapping (graph edges)	1,018

Schedule	Service timetable rules	500
Vehicle	Rolling stock and bus fleet	120

Operations

Entity	Description	Records
Trip	Individual revenue service runs	1,500
Stop Event	Real-time arrival/departure per station	3,000
Validation	Ticket tap-in events	2,000
Incident	Network disruptions and events	300
Maintenance Task	Vehicle and station maintenance	200

Analytics

Entity	Description	Records
Passenger Survey	Satisfaction ratings and feedback	400
Accessibility Equipment	Station accessibility infrastructure	180
Traffic Daily	Line-level daily ridership figures	500

Total: ~10,063 records across 16 tables

Challenge 1: Multi-Operator Data Harmonisation

With 20 operators using different IT systems, data formats vary significantly:

Issue	Impact	Mitigation
Inconsistent station naming	Duplicate/ambiguous station references	Master station registry with canonical IDs
Different vehicle ID schemes	Cannot track fleet across operators	Unified vehicle registry

Varying schedule formats	Incompatible timetable data	GTFS standard as common format
Non-standard incident coding	Inconsistent disruption reporting	Shared incident taxonomy

Challenge 2: Real-Time vs. Historical Integration

Data Type	Latency	Volume	System
Validations	Near-real-time	8M/day	AFC (Automatic Fare Collection)
Stop events	Real-time (10s)	500K/day	SIEL / SAE
Traffic daily	Next-day batch	55 records/day	BI platform
Incidents	Real-time	50–100/day	Operations centre
Surveys	Quarterly batch	400/quarter	Survey platform

Challenge 3: Network Graph Complexity

The line-station relationship creates a complex graph with:

- **Multi-line stations** — Hubs served by 5+ lines (e.g., Châtelet)
- **Branching lines** — RER lines with multiple branches (e.g., RER B: CDG ↔ Robinson + Saint-Rémy)
- **Temporal variation** — Service patterns change by day type and time of day
- **Transfers** — Physical connections not always captured in logical line-station data

Challenge 4: Unified Passenger Journey

A complete passenger journey may span:

```
Validation (entry) → Trip 1 (Metro) → Transfer → Trip 2 (RER) → Validation (exit)
```

Linking these events into a single journey requires matching card IDs, timestamps, and station-level transfer logic — a key use case for knowledge graph construction.

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6. Key Performance Indicators

Operational KPIs

KPI	Target	Measurement
Punctuality (metro)	> 96%	Trains within 2 min of schedule
Punctuality (RER)	> 90%	Trains within 5 min of schedule
Punctuality (bus)	> 85%	Buses within 5 min of schedule
Service delivery rate	> 98%	% of planned trips actually run
Incident resolution time	< 60 min	Average time to resolve disruptions
Fleet availability	> 95%	Vehicles available for service

Customer KPIs

KPI	Target	Measurement
Overall satisfaction	> 7.0/10	Passenger survey average
Information quality	> 6.5/10	Real-time display and announcement rating
Accessibility availability	> 97%	Equipment operational rate
Complaint resolution time	< 10 days	Customer service response time

Financial KPIs

KPI	Target	Measurement
Farebox recovery ratio	> 40%	Fare revenue / operating cost
Revenue per trip	> €1.50	Total fare revenue / total trips
Maintenance cost efficiency	< €0.15/veh-km	Total maintenance spend / vehicle-km
Fraud rate	< 5%	Estimated non-validated journeys

7. Appendix: Entity Relationship Overview

Primary Relationships

```
Transport Operator ──(1:N)──► Line
Transport Mode ──(1:N)──► Line
Line ──(M:N)──► Station  [via Line Station]
Line ──(1:N)──► Schedule
Line ──(1:N)──► Trip
Line ──(1:N)──► Incident
Line ──(1:N)──► Traffic Daily
Trip ──(N:1)──► Vehicle
Trip ──(1:N)──► Stop Event
Stop Event ──(N:1)──► Station
Station ──(1:N)──► Validation
Station ──(1:N)──► Accessibility Equipment
Station ──(1:N)──► Incident
Station ──(1:N)──► Passenger Survey
Ticket Type ──(1:N)──► Validation
Vehicle ──(1:N)──► Maintenance Task
Station ──(1:N)──► Maintenance Task
```

End-to-End Data Flow

```
Operator → Line → Schedule → Trip → Stop Event
                                        ↓
                                    Station
                                       │
                          Validation  Incident  Accessibility
                              ↓
                         Ticket Type

Line → Traffic Daily (aggregated ridership)
Line → Passenger Survey (quality feedback)
Vehicle → Trip (assignment)
Vehicle → Maintenance Task (lifecycle)
```

Cross-Entity Analytics

- **Passenger Journey:** Validation → Station → Line Station → Line → Trip → Stop Event
- **Network Performance:** Line → Trip (completion rate) → Stop Event (delays) → Incident (disruptions)
- **Fleet Lifecycle:** Vehicle → Trip (utilisation) → Maintenance Task (costs) → Incident (breakdowns)
- Station → Validation (traffic) + Incident (disruptions) + Equipment (accessibility) + Survey (satisfaction)
- Ticket Type → Validation → Station → Line (modal revenue share)

This document is Part 3 of the TransportIDF business description series. Together with Part 1 (Network Infrastructure & Operations) and Part 2 (Ticketing, Ridership & Quality of Service), it provides a complete business description for ontology mapping and knowledge graph construction using OntoBricks.